THE ZERO PARADOX

ZP-H Addendum

The Snap Floor in Native Categories

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ZP-H derives the Binary Snap across four mathematical frameworks and locates the snap floor ⊥ as the categorical bottom in each. In the main document that location is carried by three depth-index proxy categories — copies of ℕ with the order ≤ (Q_2BallDepth, InfoDepth, HilbDimDepth), engineered so that depth 0 is the initial object. This addendum does the stronger thing: it drops the snap floor into each framework's own category, taken directly from Mathlib, where the floor appears in that field's native syntax. A topologist, a functional analyst, and a probabilist each meet the floor in their own category and their own language, in the role that category supports: the zero object, initial and terminal, in ModuleCat ℂ (fD_zero_isInitial, fD_zero_isTerminal); an initial object that is not terminal in the Kleisli category of PMF (fC_zero_isInitial, kleisli_bottom_not_terminal); in TopCat the inverse limit of the ball tower (floorConeIsLimit, ZeroParadox/Order/PadicLimitCone.lean), which is terminal there and not initial (padic_bottom_not_initial). The three positions are stated together as fork_positions_distinct (ZeroParadox/Category/TreeSeam.lean). They are members of the bottom family (MC-1), and not one object, since an equation across categories is ill-typed. mc1_correspondence bundles the two initial objects, fC_no_return (no morphism into the floor from any later stage of the functor), and the set equation in Q_2 fB_bottom_is_limit; floorConeIsLimit sits outside it.

Three genuine Lean functors realize the snap chain inside the standard categories: F_B into TopCat (topological spaces), F_D into ModuleCat ℂ (ℂ-modules), and F_C into the Kleisli category of the probability monad PMF (stochastic maps). Each is sorry-free. F_D and F_C send depth 0 to an initial object of their categories (fD_zero_isInitial, fC_zero_isInitial), and F_D's is terminal as well, a zero object (fD_zero_isTerminal); for F_B the floor {0} is the inverse limit of its ball tower in TopCat (floorConeIsLimit), the limit of the functor's diagram rather than an object in its image. The bundled witness is mc1_correspondence (MC1Bridge.lean); its F_B field is the set equation in Q_2 beneath that limit (fB_bottom_is_limit).

Two honesty boundaries hold throughout. The realized claim is correspondence, not identity. What is proved is, category by category, the universal property named above, the floors agreeing on the snap; that the four bottoms are numerically one object is retired as ill-typed (object equality across categories does not typecheck and is not invariant under equivalence); what separates the members is proved property by property (seam_unique_among_named, for the named bottoms, in a lattice with no top). The native categories are the honest target. Mathlib has no bespoke category of p-adic spaces, Hilbert spaces, or information spaces, so the general-purpose categories — topological spaces, ℂ-modules, stochastic maps — are where the floor genuinely lives.

Section I: From Depth Proxies to Native Categories

The proxy categories of the main document are correct but indirect. Q_2BallDepth, InfoDepth, and HilbDimDepth are each a copy of $\mathbb{N}$ with the order $\leq$, wrapped so that the bottom 0 satisfies the universal property of an initial object and the source-asymmetry condition AX-G2 (no morphism returns to 0). They witness the snap floor's categorical role, but inside a structure built for the purpose rather than inside the domain itself.

The native categories are not built for the purpose, and that is the point — they also do not satisfy the ZP categorical axioms. A ZPCategory (ZP-G) requires that no terminal object exist (AX-G1): the snap has nowhere higher to go. TopCat, ModuleCat $\mathbb{C}$ and the Kleisli category all have a terminal object (the one-point space; the zero module; the one-point type, on which the only distribution is the point mass — the example after kleisli_bottom_not_terminal in ZeroParadox/Category/TreeSeam.lean). Of the categories on this page, AX-G1 holds only in the three depth-proxy categories, where the depth index has no maximum. So the snap floor cannot be "the initial object of a ZPCategory" here. The honest statement is the one each category supports natively: in TopCat the floor is the limit of the shrinking system, in ModuleCat $\mathbb{C}$ it is the zero object (initial and terminal), and in the Kleisli category it is initial and not terminal. Each is stated below.

Section II: F_B — The Snap Floor in TopCat

The 2-adic snap floor is $0 \in Q_2$. Around it sits the descending tower of clopen balls $B(0, 2^{-n})$ — the unit ball, then radius 1/2, then 1/4, and so on — each a genuine topological subspace of Q_2. The inclusions $B(0, 2^{-m}) \subseteq B(0, 2^{-n})$ for $n \leq m$ are continuous, so the tower is a functor out of $\mathbb{N}^{op}$ (the balls shrink as the index grows): a real diagram in TopCat.

Functor: fB_functor : ℕ^op → TopCat (TopFunctor.lean)
Object $n \mapsto$ the clopen ball $B(0, 2^{-n}) \subseteq Q_2$, as a TopCat object.
Morphism ($n \leq m$): the continuous inclusion of the smaller ball into the larger.
fB_bottom_is_limit: $\bigcap_n B(0, 2^{-n}) = \{0\}$
The one point common to every ball is 0: a set equation in Q_2.
floorConeIsLimit (ZeroParadox/Order/PadicLimitCone.lean): the floor {0}, with the inclusions into each ball, is a limit cone, so the snap floor is the inverse limit of the tower in TopCat; fB_limit_iso_floor identifies Mathlib's limit object with it.
Sorry-free. Purity [propext, Classical.choice, Quot.sound].

TopCat has both a terminal object (the one-point space) and an initial object (the empty space). The snap floor {0} is a one-point space, so it is terminal (padic_bottom_isTerminal, ZeroParadox/Multihomed/TreeObstructions.lean) and not initial (padic_bottom_not_initial). It is also the limit of the ball tower (floorConeIsLimit) — the categorical expression a topologist would expect for "the

point everything converges to." This is the honest realization in a category that is not a ZPCategory: not an initial object, but a limit.

Section III: F_D — The Snap Floor in ModuleCat ℂ

The state layer represents the snap in a complex Hilbert space: StateSpace n is the n-dimensional space $\mathbb{C}^n$, with the snap an orthogonal shift between basis states (ZP-D). As ℂ-modules, these spaces and the linear maps between them form a diagram in ModuleCat ℂ.

Functor: fD_functor : ℕ → ModuleCat ℂ (HilbFunctor.lean)
Object n ↦ StateSpace n = the ℂ-module $\mathbb{C}^n$.
Morphism (n ≤ m): the isometric ℂ-linear embedding that pads the extra coordinates with zero.
fD_zero_isInitial: StateSpace 0 is the initial object of ModuleCat ℂ.
The snap floor is the zero module — the one-element ℂ-module {0}, which is the categorical zero object (both initial and terminal).
fD_embed_inner: the embeddings preserve the inner product (genuine isometries).
Sorry-free. Purity [propext, Classical.choice, Quot.sound].

Here, unlike TopCat, the floor is the initial object outright: the zero object of ModuleCat ℂ is StateSpace 0. The isometry lemma records what ModuleCat ℂ forgets — that the embeddings are not merely linear but inner-product preserving — so the Hilbert-space structure travels alongside the functor rather than being discarded by the choice of category.

Section IV: F_C — The Snap Floor in KleisliCat PMF

The information layer measures the snap as one bit of divergence between distributions (ZP-C, JSD = log 2). Its native category is the Kleisli category of the probability-mass-function monad PMF: its objects are types, finite or not (PMF is defined on any type), and a morphism A → B is a stochastic map (a Markov kernel), assigning to each a ∈ A a probability distribution on B; the functor uses only finite ones, the types Fin n. Mathlib supplies this category for free, since PMF is a lawful monad; it is a Markov category in the sense of Fritz (T. Fritz, A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics, Adv. Math. 370 (2020) 107239, arXiv:1908.07021, Cor. 3.2 and fn. 10).

Functor: fC_functor : ℕ → KleisliCat PMF (InfoFunctor.lean)
Object n ↦ Fin n, the type of n distinguishable outcomes.
Morphism (n ≤ m): the deterministic embedding (each outcome mapped to its image with certainty).

Functor: fC_functor : ℕ → KleisliCat PMF (InfoFunctor.lean)
fC_zero_isInitial: Fin 0 (the empty type) is the initial object.
fC_no_return: for n > 0 there is NO stochastic map Fin n → Fin 0.
AX-G2 as a theorem: no probability distribution lives on the empty type, so no later stage returns to the floor.
Sorry-free. Purity [propext, Classical.choice, Quot.sound].

The empty type is the snap floor here, and it carries the snap's irreversibility for free. A stochastic map into Fin 0 would be a probability distribution on the empty type, and there is none. So while every object has a unique map out of Fin 0 (it is initial), no later stage Fin n with n > 0 has any map back — AX-G2 (source asymmetry) is not assumed but proved, as fC_no_return. Of the three native categories, this is the one where the framework's irreversibility axiom becomes a theorem of the ambient mathematics.

Section V: The MC-1 Correspondence, Scope, and Purity

The fields below are bundled as a single witness, mc1_correspondence; the TopCat limit cone floorConeIsLimit is named beside them, outside the bundle, and F_A (the join-semilattice ℕ of the main document, where 0 is already the initial object) is not a field of it.

Definition: mc1_correspondence : MC1Correspondence (MC1Bridge.lean)
A record collecting the native-category realizations:
• F_D: StateSpace 0 is initial in ModuleCat ℂ; it is terminal as well, the zero object (fD_zero_isTerminal, outside the bundle).
• F_C: Fin 0 is initial in KleisliCat PMF, with no morphism into it from any later stage of the functor (fC_no_return).
• F_B: the clopen balls meet in exactly {0}, a set equation in Q_2 (fB_bottom_is_limit); the limit cone in TopCat is floorConeIsLimit, outside the bundle.
Sorry-free. Purity [propext, Classical.choice, Quot.sound].

This settles the correspondence half of MC-1: across the native categories, the snap floor is realized by each category's own universal property (the zero object in ModuleCat ℂ; initial and not terminal in the Kleisli category; in TopCat the inverse limit of the ball tower, and terminal), and the four agree on the snap. The identity half, that the algebraic ⊥, the 2-adic 0, the zero module, and the empty type are numerically one object, is retired as ill-typed (object equality across categories does not typecheck and is not invariant under equivalence); what separates the members is proved property by property (seam_unique_among_named, for the named bottoms, in a lattice with no top).

Scope and Purity

This is a formalisation milestone, not new category theory. TopCat, ModuleCat $\mathbb{C}$, and the Kleisli category of PMF are standard Mathlib categories with standard universal properties. The contribution is realizing the ZP snap floor inside each as a genuine functor, in place of the $\mathbb{N}$-indexed depth proxies of the main document.

Purity: [propext, Classical.choice, Quot.sound]. Unlike the choice-free ZP-J results, these functors inherit Classical.choice from Mathlib's topology, inner-product, and PMF libraries. The dependency is a library inheritance, not a new commitment of the construction; whether it is structurally forced is the open question tracked under Classical.choice necessity (ZP-L / ZP-M).

Lean Source Files

TopFunctor.lean — fB_functor, fB_bottom_is_limit (F_B into TopCat).

PadicLimitCone.lean — floorCone, floorConeIsLimit, fB_limit_iso_floor (the TopCat limit cone).

HilbFunctor.lean — fD_functor, fD_zero_isInitial, fD_embed_inner (F_D into ModuleCat $\mathbb{C}$).

InfoFunctor.lean — fC_functor, fC_zero_isInitial, fC_no_return (F_C into KleisliCat PMF).

MC1Bridge.lean — mc1_correspondence (the bundled witness).

TreeSeam.lean — fork_positions_distinct, kleisli_bottom_not_terminal (the three positions); TreeObstructions.lean — padic_bottom_isTerminal, padic_bottom_not_initial, fD_zero_isTerminal.

All in ZeroParadox/ in the public repository.

Endnote: This document is an addendum to ZP-H Categorical Bridge and reads after it. ZP-H locates the snap floor as the categorical bottom across four frameworks using depth-index proxy categories; this addendum realizes the floor inside each framework's native Mathlib category by that category's own universal property — the inverse limit of the ball tower in TopCat (floorConeIsLimit, outside the bundle), the zero object in ModuleCat $\mathbb{C}$, and an initial object that is not terminal in the Kleisli category of distributions (fork_positions_distinct); mc1_correspondence bundles the two initial objects, fC_no_return (no morphism into the floor from a later stage), and the set equation in Q_2 fB_bottom_is_limit; floorConeIsLimit sits outside it. The floors are not one object, since an equation across categories is ill-typed. All results sorry-free in Lean 4 as of November 2026, footprint [propext, Classical.choice, Quot.sound].